

Case Study Rubric

DS 4002 – Spring 2026 - Praggnya Kanungo

Due: TBD

General Description: Please submit a link to your GitHub repository to our class on UVA Canvas.

Why am I doing this? This case study provides a way for you to use CNNs to create an image classifier for tree images to be classified into species. This project will expose you to the major parts of a data science project involving images. You will learn to visualize the exploratory analysis of a huge real-world dataset, then the analysis phase using CNNs, evaluate model accuracy, and continue to modify model architecture until you reach the desired accuracy. This will give you a lot of experience with image-classifying machine learning models.

What am I going to do? Firstly, you can find the GitHub repository for this case study at this link: https://github.com/praggnyaKanungo/DS4002_CS3_Tree_Case_Study/tree/main Using this link, you will have everything you need in order to land your deliverable for this project. In the DATA directory in the GitHub repository, you will find the dataset of images for you to use in your CNN-based classifier. Your goal is to create a classifier and to keep tweaking the model architecture until you land a high accuracy. After you are done with this, you will see if your model can be generalized to real images of Charlottesville-based trees by testing if your model can accurately classify images of trees you have taken yourself.

Your final deliverable will have the following:

- At least 4 visualizations for the exploratory data analysis of the initial image dataset (and a clearly commented source code for generating these plots)
- A trained tree image to species classifier using CNNs (and clearly commented source code for this)
- An executed method for model accuracy evaluation (elaborate on this process in your README.md)
- Use of your classifier to predict the species of new tree images taken in Charlottesville (elaborate on this also in the README.md)

How will I know I have Succeeded?

If you follow the below rubric, you will definitely succeed! Please just read the rubric carefully and ensure you provide all requirements.

Formatting	All you need to do is submit a link to your GitHub Repository that has all of the following: - o A DATA directory with the DATA you used for this project (which is already provided in the Case Study Github) ▪ Inside this directory, please have Tree_Photos directory where you will put your pictures taken of Charlottesville Trees - o A SCRIPTS directory with files of course code for all the phases of the project including: ▪ A file that generated the visualizations for the exploratory analysis ▪ A file training the classifier of images of trees

	- o A OUTPUT directory where all your outputs generated from the scripts will go. For example, the plots you generate should go here (you must have at least 4 plots). - o LICENSE (For example, you may use a MIT license) - o README.md ▪ This should summarize what you did in this project as well as your results.
README.md	• <u>Goal</u>: Anyone who visits your repository and sees the README should have a clear idea of your project repository, what you did in this project, as well as your results. • The way you want to format your README.md is up to you, but you must provide the following information: - o Explain the structure of your repository - o Explain what the purpose of your project was - o Explain how you did the project - o Explain your results in terms of model accuracy and how you evaluated that - o Explain how your model performed on pictures of Charlottesville Trees you have taken - o List your references and any links you used in order to write your source code or README. For more information about this, please see the References part of this rubric.
Code	• <u>Goal</u>: Create course code that functions appropriately and ensures that anyone who reads it is able to understand it. • You are allowed to use multiple files, and you are encouraged to do so. For example, you may use a file per phase of the project sure as: - o One for the exploratory plots - o One for the creation of the model classifier - o One for testing your model on your personally taken images • All your files must be in the SCRIPTS directory. • Comment all your code files so anyone who looks at them is able to easily identify different functionalities of your different parts of code.
References	• <u>Goal</u>: Provide all the references you used to create this project. • At the end of your README, please list all your references in IEEE citation style (only include the links you used).

Acknowledgements: Thank you to Professor Rasero, whose rubrics I based the structure of this rubric on!